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NEURAL BELLMAN OPERATORS

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Neural Bellman Operators construct and learn continuation operators for feasible decisions under an original recursive economic objective. We derive an action-scaled comparison modulus for the constrained investment economy with continuous innovations and nested average value at risk. Intersecting paired continuation differences with convex value bounds yields a computable action-aligned certificate. Matching information bounds hold for a stated fixed-risk-density point-oracle class; a constrained tail-defect transfer and the implemented future-policy allowance extend the calculation to actual policy changes. We prove uniformly safe replacement, finite-resolution sufficient conditions and the remaining future-policy information floor. The stated Bellman, actor–critic, recursive-utility and boundary propositions receive complete proofs under their separate assumptions. Eight representations with common access to the original primitives and four complete services are compared under a separately frozen independent protocol, with all acquisition, refusal, qualification and service costs. Independent rational reconstruction supplies original-policy and global-optimum bounds. All 42 learned-incumbent refinement groups obtain positive simultaneous mean-gain lower bounds for the declared state population. The new mixed bound authorizes one shared candidate at a recorded resolution where the separate convex bound is inconclusive, while the paired service is slower than inherited safe improvement in every matched clock. Strong online and direct conventional methods remain competitive; no neural advantage is established against the complete conventional menu. The results identify the joint learning and certification work required for economically valuable NBO decisions.

KEYWORDS: Dynamic programming, Neural approximation, Recursive risk, Policy qualification, Computational economics.

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1. INTRODUCTION

An economic decision rule must choose a feasible action using the information available at the decision date. Its numerical value is determined by the resulting recursive economic objective and by the resources needed to deliver that action. Neural Bellman Operators (NBOs) study the construction and learning of continuation operators for this purpose. A learned critic represents the continuation entering a Bellman problem; an actor obtains an admissible decision from that represented problem; and the value of the implemented policy follows its own future decisions under the original economic recursion. This distinction becomes consequential with nonlinear recursive risk, state-dependent action constraints, imperfect optimization, and finite information acquisition.

The central question is when learning a continuation operator improves economic decisions at a competitive complete cost. Answering it requires more than a function-approximation result. A small fitted value residual can coexist with a poor action direction, an incorrect boundary decision, an expensive risk evaluation, or an uncertified future policy. A representation that is cheap to query can also be expensive to acquire or to qualify. The original NBO framework therefore keeps the target Bellman recursion, the implemented policy recursion, and the computational account explicit. This revision advances that framework through action comparisons whose information requirements depend on the actions being compared, followed by a new original-objective policy-improvement study.

Our principal application is the original constrained investment economy. A firm has a vector of productive states, a state-dependent investment capacity, a nonlinear propagation technology, and continuous innovations. Current and terminal costs penalize distance from productive targets, cross-component imbalances, and severe aggregate shortfalls. Upper-tail cost risk is evaluated recursively, so the adverse continuation distribution affects the investment decision at each date. The model has a uniquely defined original Bellman optimum and a measurable implemented terminal controller. These objects permit economic comparisons of neural, hinge, polynomial, spline, grid and semi-analytic continuations using one original objective and feasible action set, with common access to the economic primitives.

The application is a controlled economic numerical experiment; its cost units and initial-state distribution are declared rather than estimated from empirical firm data.

An action-scaled Bellman comparison. The first contribution is a quantitative comparison modulus derived from the original constrained technology and recursive preference. Let $f_a(z) = V_1(F(x,a,z))$ denote the original optimal continuation after action a. For the two-dimensional investment problem we prove a global Lipschitz bound M_α for the state gradient of V_1, including binding investment constraints and changes of the active risk tail. Normalizing the feasible action interval makes the boundary calculation explicit. It follows that $\mathrm{Lip}(f_b - f_a) \le (4/5)M_\alpha|b-a|$. Thus the fixed-density integration error of the continuation difference admits a bound proportional to action distance, even when the level of each continuation remains substantial.

This modulus gives a computable paired midpoint interval. Intersecting it with the separately valid convex midpoint–secant intervals yields the integration width

$$\min\{2KR(1-\alpha)n^{-2},\ E_{ab}R(1-\alpha)n^{-1}\}, \qquad E_{ab} = (4/5)M_\alpha|b-a|,$$

under a known feasible risk-tail density and exact point information. Acquired node errors, numerical accumulation, tail localization and future-policy error are then charged explicitly. We prove a matching worst-case order for a precise fixed-density scalar-value information interface. The comparison applies to all representations given that interface. The additional smoothness and algebraic access available in the investment economy are stated separately; the abstract lower bound does not remove these advantages from a conventional solver.

The second contribution connects action alignment to the movement of the adverse tail. A common feasible density has a nonnegative deficit relative to the maximizing density of each continuation. Strong convexity in the innovation and the action-scaled comparison modulus transfer a bound for one deficit to the other, including a tail optimizer at the boundary. At an exactly optimal candidate tail, the resulting mismatch allowance is quadratic in action distance. The implemented

comparison keeps both directly acquired deficits and intersects the incumbent deficit with this transfer bound. It retains the original risk supremum throughout.

A policy change must improve the implemented policy. The new gate starts from a completed, qualified incumbent. A candidate is accepted only when its original complete-policy contrast has a strictly negative upper bound and the candidate satisfies the unchanged original regret and interval-width requirements. Every accepted step therefore has a certified economic gain. Every refusal, unresolved comparison, and proposal acquisition remains in the cost account. The three projected support proposals and the completed-incumbent safeguard are retained methods; the new numerical mechanism changes the information used to authorize their realized policy changes.

We prove a finite-resolution sufficient condition for acceptance, with distinct allowances for common-tail mismatch, point accuracy, integration, and actual future behavior. There is also a sharp limit to the stated information: knowing the optimal-future action values exactly leaves a comparison uncertainty of $\pm\beta\varepsilon_1$ when the implemented future policy is known only to have error in $[0, \varepsilon_1]$. Resolving smaller gains requires information about that actual future policy. This result identifies a specific numerical research problem within NBO rather than treating an unresolved comparison as optimality or interpreting additional internal fitting accuracy as economic progress.

Evidence in the original economic units. The new experiment fixes eight continuation representations and four complete services before independent initial states are generated. In addition to the completed incumbent and the new paired service, it includes the retained three-trial safe method and a stronger direct original-continuation proposal that receives the same paid comparison gate. Every learned representation uses the same original V_1 value labels. The hinge control receives the neural features and pays their acquisition in full. The semi-analytic original solver, quadratic and polynomial controls, coordinate spline, and conventional grid remain eligible competitors. Actual information counts and complete costs are reported even when they differ across methods.

The independently generated 64 state rows support 4,096 primary decisions. Three additional serial processes repeat a fixed 32-state subset for descriptive timing, without creating additional independent observations. A separate exact rational implementation reconstructs every unique returned state/action/risk key. It evaluates the original continuous terminal risk and also constructs an all-action lower bound for the original date-zero optimum. This gives original policy regret and action-distance bounds in addition to fixed-policy value enclosures. Prespecified binomial and bounded-mean inference transfer certified gains over each service's own incumbent to the declared state population. Cross-representation adoption still requires comparing complete economic costs and all acquisition and decision-service resources.

Relation to numerical control and learning. Monotone improvement from approximate policy evaluation has a substantial literature. Metelli et al. (2021) construct policy-improvement lower bounds and safe policy-iteration schemes. Eshwar et al. (2026) study Bellman-constrained evaluation and conservative improvement with function approximation. These works make the general need for an improvement guarantee clear. The present addition concerns the paid continuous-risk action-comparison problem generated by the original NBO economy, its explicit constrained smoothness and tail-mismatch constants, and the policy consequences of that information. Accuracy certificates for convex minimization with inexact oracles also supply valid stopping criteria (Gladin et al., 2023). Here the economic recursion requires separately accounting for continuation, risk, feasible actions, and the actual future controller; an internal optimization residual alone does not provide those accounts. Classical binomial inversion and concentration are attributed when used, and ordinary hinge or polynomial expressiveness is not credited to neural networks alone.

The main article presents the NBO construction, original recursive-risk account, investment model, new action comparison and economic evidence. The Technical Supplement proves the Bellman foundations, stability, own-future errors, constrained gradients, full critic–actor graph, information statements and new results. It also retains the separate Brownian recursive-utility and stopped boundary bridge

under its own assumptions. The Development Appendix preserves the broader economic program, previous theorems, complete historical comparisons, failed or adverse results, and source-specific responses. Historical experiments retain their own protocols and evidence identities; the new independent sample is used only for its prospectively declared study.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the earlier expected-cost experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. Section 5 evaluates the same investment technology under an explicitly nested AVaR preference and retains the original continuous innovation law. The direct-cost inference in Section BN is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1}, returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the

first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \le \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^\pi f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^\pi - V_0^* \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 1.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \tag{5}$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in* (5) *have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \le f_t - \mathcal{T}_t f_{t+1} \le D_t k_t + e_t + 2L_t h_t, \tag{6}$$
$$\mathcal{T}_t^\pi f_{t+1} - f_t \le e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \le \sum_{t<T} B_t (D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T (2e_T + 2L_g h_T). \tag{7}$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite

construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 1.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

4. REFERENCE-CENTERED RISK KERNELS AND COMPLETE-POLICY QUALIFICATION

A datewise worst-case error account can charge incompatible states at successive dates. Under nonlinear recursive preferences, replacing that account by an expectation under the physical transition law is not valid. We instead retain the conditional risk evaluation and center its dual representation at the actual stored NBO continuation. The resulting state-dependent certificate has an original-optimum interpretation and a finite occupation representation. It extends, rather than replaces, the signed qualification of Section AJ.

4.1. *The centered operator*

Retain the original augmented economic state, feasible actions, nonnegative discounts, and bounded complete-policy recursions. At each (t, x, a) suppose the original conditional evaluation admits the representation

$$\rho_{t,x,a}(v) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int v(y) q(dy) - \psi_{t,x,a}(q) \right\}. \tag{8}$$

Every q is a probability measure on the original successor state space. The representation is assumed on a common bounded-function domain containing all continuations below, with the measurability and selection conditions needed for the original Bellman recursions. This is a hypothesis about the original preference

primitive, not permission to replace a static risk objective by a nested one. Conditional risk duality and risk-averse dynamic programming are established principles (Ruszczyński, 2010, 2014). For a nonconvex or non-cash-invariant aggregator lacking (8), the earlier order and modulus results remain applicable.

Fix one qualified parameter θ and suppress its subscript. Its entire sequence of stored continuations is $f = (f_t)_{t=0}^{T}$, and its supplied complete feasible policy is π. Define, using the same f_{t+1} in every term,

$$\begin{aligned} r_t(x,a) &= c_t(x,a) + \beta_t \rho_{t,x,a}(f_{t+1}) - f_t(x), \\ k_t^f(x,a,q) &= \psi_{t,x,a}(q) + \rho_{t,x,a}(f_{t+1}) - \int f_{t+1}\, dq, \\ \mathcal{E}_{t,x,a}^f(z) &= \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int z\, dq - k_t^f(x,a,q) \right\}. \end{aligned} \tag{9}$$

The penalty is nonnegative and has infimum zero, whether or not the supremum in (8) is attained. Direct substitution gives

$$\mathcal{E}_{t,x,a}^f(z) = \rho_{t,x,a}(f_{t+1} + z) - \rho_{t,x,a}(f_{t+1}). \tag{10}$$

In particular, $\mathcal{E}^f(0) = 0$, and $\mathcal{E}^f$ is monotone, cash invariant, and nonexpansive in the uniform norm. Discarding k^f is an upper relaxation, not this identity. Freezing one risk-maximizing kernel at f is in general not even an upper relaxation.

THEOREM 3—State-dependent original-optimum certificate: *Suppose verified functions satisfy $r_t^-(x,a) \le r_t(x,a)$ for every original feasible action and $r_t(x,\pi_t(x)) \le r_t^{\pi,+}(x)$. Let bounded measurable L_t, U_t satisfy*

$$\begin{aligned} L_T &\le g - f_T \le U_T, \\ L_t(x) &\le \inf_{a \in A_t(x)} \{ r_t^-(x,a) + \beta_t \mathcal{E}_{t,x,a}^f(L_{t+1}) \}, \\ U_t(x) &\ge r_t^{\pi,+}(x) + \beta_t \mathcal{E}_{t,x,\pi_t(x)}^f(U_{t+1}). \end{aligned} \tag{11}$$

Then, at every initial state,

$$f_0 + L_0 \le V_0^* \le V_0^\pi \le f_0 + U_0, \qquad 0 \le V_0^\pi - V_0^* \le U_0 - L_0. \tag{12}$$

The same statements hold at every intervening date and state. They are uniform over a qualified parameter class whenever all premises and the reported bound hold for each member with its own common continuation sequence and complete policy. The certificate permits endogenous states and actions changing at every date; it does not require a finite state space or independent deployment contexts.

With exact residuals and terminal discrepancies, equality recursions in (11) *give* $L_t = V_t^* - f_t$ *and* $U_t = V_t^\pi - f_t$. *With scalar bounds from* (A.34), *their corresponding constant lower and upper sequences are feasible certificates here. Thus the exact state-dependent recursion is no looser than the signed scalar account. A strictly smaller bound can result without changing a fit, a feasible action, or an economic observation.*

The last assertion concerns the best exact recursion or a verified improvement upon the constant certificates. It is not a claim that every approximately computed L, U improves the scalar account. In implementation, the minimum of valid policy supersolutions and the maximum of valid optimum subsolutions retain validity, so an inferior new bound need not displace the existing one.

The lower inequality requires every original action. A discrete action search without a verified original-menu lower bound cannot establish it. For a fixed action, a feasible risk kernel supplies a lower bound on $\mathcal{E}^f(L)$; an upper bound requires control of the entire risk envelope. Numerical integration, feasible repairs, residual errors and penalty errors belong in these inequalities. If penalty enclosures are used, lower penalty endpoints enter an upper certificate and upper endpoints enter a lower witness. Approximate dual optimizers are accepted only through these independently checked inequalities.

5. CONTINUOUS-LAW QUALIFICATION IN THE INVESTMENT MODEL

The risk theorem becomes a numerical method only when its inequalities can be constructed from economic primitives. We now supply this construction for the original investment technology with a nested tail-risk preference. States and innovations remain continuous, every feasible action is covered, and the implemented terminal controller rather than a hypothetical exact optimizer determines policy cost. This application joins continuation construction, the original-optimum

comparison and a representation comparison on one problem. It does not replace the general NBO construction or the other economic applications.

5.1. *Primitives and the object being approximated*

Let $d \geq 2$ be even, $X = [0,1]^d$, and interpret subscripts cyclically. At dates $t = 0, 1$, the feasible investment is $0 \leq a \leq A(x) = 1/8 + \bar{x}/8$, where $\bar{x} = d^{-1} \sum_j x_j$. Write $b_j = 1/2$ for odd j and $b_j = 1/4$ for even j, and $s_j = (-1)^{j+1}$. The original transition is

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + b_j a + s_j z, \qquad z \sim \mathrm{Unif}[-1/32, 1/32]. \tag{13}$$

Define

$$S(x) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2(1/2 - 2\bar{x})_+^2, \tag{14}$$

$$g(x) = S(x) + \frac{2}{d} \sum_j (x_j - 5/8)^2, \qquad h(a) = a^2 + 4a^4.$$

We use $\beta = 15/16$ and, separately, $\alpha \in \{1/2, 3/4\}$. Average value at risk is the upper-tail cost functional

$$\rho_\alpha(v) = \sup \left\{ \int v q \, dP : 0 \leq q \leq (1 - \alpha)^{-1}, \int q \, dP = 1 \right\}.$$

Here P is the original uniform innovation law. The objective is recursively defined by $V_2 = g$ and

$$V_t(x) = S(x) + \inf_{0 \leq a \leq A(x)} \{ h(a) + \beta \rho_\alpha(V_{t+1}(F(x, a, \cdot))) \}.$$

It is not the static tail risk of total realized cost. A complete policy value replaces each infimum by the action prescribed by that policy, with the same recursive risk primitive.

Assumptions for the continuous construction. For this subsection, the primitives are exactly (13)–(14); α is fixed before construction; the fitted continuation f is a finite convex function on X; and every numerical lower or upper value is an outward

enclosure. Feasible tail densities and feasible actions are retained explicitly. Terminal selection is the measurable, dyadically acquired controller described below, with its fallback included in the error bound. These concrete conditions imply compact feasible graphs, bounded costs, measurable minimizers, and the conditional-risk dual used in Theorem 3. They do not assert that every recursive preference or every trained network has this convex structure.

PROPOSITION 4—Primitive curvature and state supports: *For either risk level and every even d, the one-date objective $Q_1(x,a) = S(x) + h(a) + \beta\rho_\alpha(g(F(x,a,\cdot)))$ is jointly convex on $X \times [0,1/4]$, with curvature bounded below by*

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2 \tag{15}$$

in a direction (v,w). Its value V_1 is convex on X and satisfies

$$|V_1(x) - V_1(y)| \le \frac{15}{d}\|x - y\|_1. \tag{16}$$

Let W be the objective evaluated under any fixed feasible risk density, at an anchor (x_0,a_0), and let (p_x,p_a) be its gradient there. For any $\lambda,\nu \ge 0$, set $e = p_a + \lambda - \nu$. Then the following is a global affine minorant of V_1:

$$\ell(y) = W + \lambda(a_0 - A(x_0)) - \nu a_0 - \frac{e^2}{4} + \left(p_x - \frac{\lambda}{8d}\mathbf{1}\right)^\top (y - x_0). \tag{17}$$

Neither optimal multipliers nor an exact optimizing action are required.

The curvature is obtained from the original nonlinear technology, not imposed on an interpolated value grid. The negative second-order term from $x_j(1 - x_{j+1})$ is bounded using $|\partial_j g| \le 10/d$; the stage curvature dominates it. The allowance $e^2/4$ pays for imperfect stationarity. The proof in the Technical Supplement also establishes the global state modulus in (16), including the state-dependent action constraint.

5.2. *An all-action certificate under the original innovation law*

Partition the innovation interval into n equal bins B_i, with midpoints z_i and probabilities $p_i = 1/n$. An innovation bin is a set over which an inequality holds, not a replacement point mass in the economic model. Choose nonnegative $q_i \leq p_i/(1-\alpha)$ summing to one. These masses define a feasible density constant on each bin. For a fixed initial state and feasible proposal a_0, let W_0 be a lower bound on the objective under this density, obtained from globally valid affine supports of the continuation at the successor midpoints. Let p enclose the derivative of the same supporting objective at a_0. Then

$$\underline{V}_0(x) = \inf_{0 \leq u \leq A(x)} \{W_0 + p(u - a_0) + (u - a_0)^2\} \tag{18}$$

is a lower bound on the original Bellman optimum when the supported continuation is V_1. The scalar quadratic is minimized over the full original interval. Interval uncertainty in p is covered by taking the smaller minimum at its two endpoints, since the minimum is concave in p.

For a convex fitted continuation, midpoint supports give the lower risk witness and the larger endpoint value bounds the continuation throughout each bin. For V_1, Proposition 4 supplies midpoint supports and the allowance $15/(32n)$ covers each half-bin. Upper tail risk is the average of the largest $n(1-\alpha)$ upper bin bounds. This average is the exact risk of a dominating step function, hence an upper bound under the continuous law.

THEOREM 5—Constructed original-optimum qualification: *Suppose the supports and bin bounds above are outwardly verified. Suppose further that the complete terminal controller satisfies $V_1 \leq V_1^{\widehat{\pi}} \leq V_1 + \varepsilon_1$ everywhere on X. For every queried initial state, the lower value (18) and the binwise upper bound $\overline{J}_0^{\widehat{\pi}}$ satisfy*

$$0 \leq J_0^{\widehat{\pi}}(x) - V_0(x) \leq \overline{J}_0^{\widehat{\pi}}(x) - \underline{V}_0(x). \tag{19}$$

For any stored convex continuation f, the same lower and upper constructions are admissible bounds for the centered-risk comparison of Theorem 3. Their validity does not depend on the representation of f, on a successful fit, or on the proposal being an exact minimizer.

To see the connection, write a verified continuation bracket as $\ell \le V_1 \le u$ and use errors $L = \ell - f$, $U = u - f$. The centered identity gives

$$\rho_\alpha(f) + \mathcal{E}^f(L) = \rho_\alpha(\ell), \qquad \rho_\alpha(f) + \mathcal{E}^f(U) = \rho_\alpha(u).$$

Joint evaluation avoids subtracting and then adding two separately rounded copies of $\rho_\alpha(f)$. The bound (18) uses one feasible risk density to minorize the required all-action infimum; the policy upper bound controls the full risk envelope. Thus the application constructs sufficient inequalities for the original theorem. It does not claim that a non-neural continuation would receive a different validity theorem, or that computing the original-value witness is free.

The terminal controller observes each state coordinate rounded down to 40 binary places. It accepts its floating proposal only after an outward all-action gap of at most 2^{-21} at that observed state; otherwise it uses rational bisection. The original-state transfer is less than $30\,2^{-40}$. Consequently $\varepsilon_1 = 2^{-20}$ covers the complete controller, including observation and fallback. Policy evaluation encloses this controller at every possible successor, not only at the midpoint states used by the integrator.

Well-posed original decisions. The comparison uses a specified conditional nested recursion on the original observation filtration. A sufficient primitive model has a compact state space, nonempty compact feasible sets varying continuously with state, continuous costs and invariant transitions, compact innovations with their original fixed laws, and finite monotone cash-invariant risk evaluations. These assumptions make the Bellman operator preserve continuous functions. For implemented Borel policies, the risk evaluation must also preserve Borel parameter kernels; the supplement proves this property for expectation and AVaR. Proposition S.2 proves, rather than assumes, the resulting finite-horizon value existence and uniqueness, attained Borel minimizers, comparison against all feasible adapted history policies, and the dated stability bounds. These are classical foundations (Ruszczyński, 2010, 2014), not a claim of a new fixed-point principle. Pointwise transition continuity with an unbounded innovation support is insufficient for arbitrary monetary risk; the exact essential-uniform alternative is stated in the proof.

The original investment primitives satisfy these conditions directly: $F(x,a,z) \in [1/32, 27/32]^d$, $1/8 \le A(x) \le 1/4$, and all costs are bounded. Their joint curvature gives V_1 state strong-convexity modulus $(4 - 5\beta/4)/d$ and root action modulus two. Both dates therefore have unique continuous constrained optimal actors. Proposition S.3 also derives the continuous value gradient with its capacity multiplier, the original-risk root derivative, and the actual observation/feasibility allowance. Differentiating a rounded actor is not used to establish any of these claims. The supplement records the full smooth critic–actor graph and distinguishes its gradient from a frozen-target update.

The separate Brownian recursive-utility bridge has its own existence and verification assumptions. Proposition S.4 supplies the stopped-domain version, charging the original stopping boundary and terminal condition. Its Lipschitz driver can depend on the backward value, in which case cash invariance is not automatic. The continuous-time stability factor and smooth critic graph are therefore verified within that original recursive criterion.

6. ACTION-ALIGNED ACQUISITION AND COMPLETE-POLICY WORK

A continuation is acquired to make a decision, not to minimize an approximation statistic in isolation. The retained investment experiment makes the distinction operational: every certificate-oriented fit reduces its validated residual span, whereas aggregate first-proposal rejections increase. We now construct an acquisition target in the units of the admission inequality. The construction uses the same original action set and risk law as the NBO policy account.

6.1. *A computable directional certificate*

Fix a state and date, and suppress their indices. Let $A \subset \mathbb{R}^k$ be nonempty, compact and convex, let Q be the original action objective, and let $a \in A$ be the actually realized proposal. A verified supporting calculation supplies $L, U \in \mathbb{R}$, $s \in \mathbb{R}^k$ and $\mu > 0$ such that

$$Q(a) \le U, \qquad Q(b) \ge L + s^{\mathsf{T}}(b-a) + \frac{\mu}{2}\|b-a\|_2^2 \quad \text{for every } b \in A. \tag{20}$$

These inequalities need not identify s as a derivative of Q. In particular, an imperfect risk witness or continuation minorant can lower L while retaining a useful supporting slope. Put $w = U - L$ and

$$\Gamma(a) = w - \min_{b \in A} \left\{ s^{\mathsf{T}}(b-a) + \frac{\mu}{2}\|b-a\|_2^2 \right\}. \tag{21}$$

A certified lower bound on the displayed minimum gives a valid, possibly larger, Γ. An unverified optimizer status does not.

THEOREM 6—Directional certificate for an NBO proposal: *Under* (20),

$$0 \le Q(a) - \inf_{b \in A} Q(b) \le \Gamma(a).$$

Let $\widehat{s}$ be a supporting slope of the surrogate used to propose a, let $N_A(a) = \{n : n^{\mathsf{T}}(b-a) \le 0 \ (b \in A)\}$, and suppose $\|s - \widehat{s}\|_2 \le d$. If a supplied $n \in N_A(a)$ satisfies $\|\widehat{s} + n\|_2 \le r$, then

$$\Gamma(a) \le w + \frac{(d+r)^2}{2\mu}. \tag{22}$$

For $A = [0, A_0]$, the exact minimum in (21) *is attained at $b = \mathrm{proj}_{[0,A_0]}(a - s/\mu)$.*

The proof, including boundary actions, is in Supplement 5. Equation (22) separates three trainable or computable quantities: the original-objective enclosure width w, the directional mismatch d, and the surrogate stationarity residual r. The residual span of the continuation is not one of these quantities. A readout can improve that span and worsen the supporting slope at a nearly binding action. Conversely, constant changes in a continuation need not affect any of them after consistent centering.

The supporting calculation is available under the original recursive-risk primitives. Suppose $F(b,\omega) = F(a,\omega) + G(\omega)(b-a)$, the stage cost is μ-strongly convex, and the risk dual has an action-independent feasible witness q on the original innovation space. If

$$V(F(b,\omega)) \ge \ell(\omega) + v(\omega)^{\mathsf{T}} G(\omega)(b-a),$$

then (20) holds with

$$L = c(a) + \beta \left\{ \int \ell \, dq - \psi(q) \right\}, \qquad s = \nabla c(a) + \beta \int G^{\mathsf{T}} v \, dq, \tag{23}$$

and any valid upper bound U for the original risk objective. Integration and penalty errors must be enclosed in these quantities. The witness must remain feasible for every action being minorized; a witness feasible only at a is insufficient when the risk set itself depends on the action.

In the investment economy, $G = (1/2, 1/4)^{\mathsf{T}}$ at $d = 2$, the innovation law and AVaR cap are action independent, and $h'' \geq 2$. Thus $\mu = 2$ gives exactly the all-action quadratic of (18). The added result is the explicit link from an acquired directional support to its admission gap, rather than a new economic model or an assumption that an acquired value support is an exact derivative.

7. RESOLVING THE ACTIVE INNOVATION TAILS

The original continuous innovation law supplies information that an NBO must share with its strongest conventional competitors. Convexity of a continuation along that innovation implies that the adverse outcomes occupy at most two endpoint intervals. Locating these intervals before quadrature changes which continuation values must be acquired. In particular, the upper information bound need not pay a factor $(1 - \alpha)^{-1}$ for resolving the entire innovation support when only a fraction $1 - \alpha$ is adverse. The result below is a deterministic point-information theorem and an outward Bellman implementation. It does not assert a neural representation separation, a randomized optimum, or multivariate tractability.

7.1. *A tail-uniform point-information account*

Let $f : [-R, R] \to \mathbb{R}$ be convex and K-Lipschitz, where $R, K > 0$; the information consists of exact point values, the original uniform law, and these class restrictions. The formula of f, its derivatives, and its integrals are not supplied. Write $t = 1 - \alpha$ and $L = 2R\alpha$ for $0 < \alpha < 1$, and let $I = [-R, R - L]$. For $u \in I$ put

$$H(u) = \int_u^{u+L} f(z) \, dz, \qquad D(u) = f(u + L) - f(u), \tag{24}$$

$$R_u(f) = \frac{\int_{-R}^{u} f(z)\,dz + \int_{u+L}^{R} f(z)\,dz}{2Rt}. \tag{25}$$

Every R_u is a feasible original-law AVaR witness: its density relative to the original uniform law is t^{-1} on the two displayed intervals and zero elsewhere. Its feasibility does not require u to be optimal.

THEOREM 7—Active-tail information with uniform risk-level constants: *The function H is convex and continuously differentiable, $H' = D$, and D is nondecreasing and $2K$-Lipschitz. Moreover,*

$$\mathrm{AVaR}_\alpha(f) = \max_{u \in I} R_u(f). \tag{26}$$

If $D(-R) \geq 0$, the left endpoint is a maximizer in (26); if $D(R-L) \leq 0$, the right endpoint is one. Otherwise a zero of D lies in the interior. After m sign bisections, choose the midpoint u_m of the retained zero bracket, terminating early at any exact zero. Then

$$0 \leq \mathrm{AVaR}_\alpha(f) - R_{u_m}(f) \leq \frac{KRt}{2}\,4^{-m}. \tag{27}$$

The right side is replaced by zero after an endpoint or exact-zero exit.

For an integer $n \geq 1$, partition each of the two selected tails into equal cells within that tail, with every cell length at most $2Rt/n$. At most $n+1$ nonempty cells are required. Midpoint values give a lower bound $\underline{R}$ for $R_{u_m}(f)$; endpoint secants give an upper bound $\overline{R}$. The resulting interval

$$\left[\underline{R},\ \overline{R} + \frac{KRt}{2}4^{-m}\right] \tag{28}$$

contains the original AVaR and has width at most

$$KRt\left(n^{-2} + \tfrac{1}{2}4^{-m}\right). \tag{29}$$

With $m = \lceil \log_2 n \rceil$, the width is at most $3KRt/(2n^2)$ and at most $2n + 2\lceil \log_2 n \rceil + 8$ distinct point values are acquired, including tail localization.

Conversely, every deterministic adaptive interval algorithm using at most N exact point values and sound for all consistent convex K-Lipschitz functions has an input on which its

reported width is at least

$$\frac{KRt}{6(N+1)^2}. \tag{30}$$

Thus, for $N \geq 32$, the optimal worst-case width is of order $KR(1-\alpha)N^{-2}$ with constants independent of $\alpha \in [0,1)$, under this exact point-information convention. At $\alpha = 0$ the construction uses ordinary midpoint–secant integration on the full interval and no localization; the same order and lower bound hold.

The lower bound is the retained indistinguishable-tail argument, with its complete proof restated in the supplement. The addition is an upper algorithm that locates the adverse interval and pays for that localization, so that the same linear tail-length factor appears on both sides. The AVaR variational principle and midpoint–secant inequalities are classical (Rockafellar and Uryasev, 2002, Novak, 2015). This result does not make a point-value oracle equivalent to formula access. As α approaches one, representing the shrinking intervals, resolving nearby function values, and enforcing outward arithmetic may require additional bits and numerical work. Those costs are outside an exact-real point-query count and are charged in the implementation.

7.2. *An outward implementation for the original investment economy*

Fix the original investment state x, a feasible realized proposal a, the same α, and the unchanged implemented terminal controller. Let

$$f_a(z) = V_1(F(x,a,z)), \qquad Q(a) = S(x) + h(a) + \beta\,\mathrm{AVaR}_\alpha(f_a).$$

Here Q uses the original optimal last-date continuation. Its relation to the actual complete-policy cost $J(a)$ is

$$Q(a) \leq J(a) \leq Q(a) + \beta\varepsilon_1, \qquad \varepsilon_1 = 2^{-20}. \tag{31}$$

This allowance is inherited from the actually implemented controller; it is not inferred from the newly chosen quadrature nodes.

The primitive curvature in Proposition 4 implies that V_1 is m_x-strongly convex, with $m_x = (4 - 5\beta/4)/d$. Since the original innovation direction has squared Euclidean length d, f_a is m_z-strongly convex, where

$$m_z = 4 - 5\beta/4 = 181/64. \tag{32}$$

Its Lipschitz constant in z is at most 15. Consequently the integrated central loss H_a is $m_z L$-strongly convex. These are original-primitive constants, not fitted shape estimates.

Let any feasible $u \in I$ be supplied, with a verified interval $[d_-, d_+]$ containing $f_a(u + L) - f_a(u)$. Define

$$\kappa(u) = \frac{1}{2Rt} \max_{d \in \{d_-, d_+\}} \left[-\min_{v \in I} \{ d(v - u) + \tfrac{1}{2} m_z L (v - u)^2 \} \right]. \tag{33}$$

The inner minimum is attained at $v = \mathrm{proj}_I(u - d/(m_z L))$; an outward lower bound on that minimum is sufficient. Thus inaccurate endpoint comparisons enlarge a retained defect instead of granting false tail optimality.

Partition the selected tails into cells $B_i = [l_i, r_i]$ with midpoint z_i and positive weight $w_i = (r_i - l_i)/(2Rt)$. Their weights sum to one. At each successor $y_i = F(x, a, z_i)$, obtain a valid global affine minorant

$$V_1(y) \geq \ell_i + p_i^{\mathsf{T}} (y - y_i).$$

The vector p_i may itself be enclosed, provided the uncertainty is retained in the final slope. Obtain upper values u_i^- and u_i^+ for $V_1(F(x, a, l_i))$ and $V_1(F(x, a, r_i))$. Put

$$r_- = \sum_i w_i \ell_i, \qquad r_+ = \sum_i w_i (u_i^- + u_i^+)/2, \tag{34}$$

$$W = S(x) + h(a) + \beta r_-, \qquad s = h'(a) + \beta \sum_i w_i b^{\mathsf{T}} p_i, \tag{35}$$

where b is the original investment action-exposure vector. All sums and products must be outward enclosures of these exact weighted objects.

THEOREM 8—Original-policy qualification from a feasible active tail: *Suppose the displayed supports, value intervals, derivative interval, and scalar minimizations are valid*

outward bounds. Then

$$S(x) + h(a) + \beta r_- \le J(a) \le S(x) + h(a) + \beta\{r_+ + \kappa(u) + \varepsilon_1\}, \tag{36}$$

$$V_0(x) \ge \inf_{0 \le v \le A(x)} \{W + s(v-a) + (v-a)^2\}. \tag{37}$$

If $s \in [s_-, s_+]$ is uncertain, taking the smaller of the two minima at s_- and s_+ gives a valid lower bound in (37). Subtracting this lower bound from the policy upper endpoint therefore bounds complete-policy regret relative to the original optimal policy. The original loss target $1/256$ and policy-width target $1/8192$ retain their original meanings.

The conclusion applies to boundary proposals, a one-sided tail, a nonoptimal tail cut, a rounded action, and an imperfect stored continuation used only to generate the proposal. It requires neither convexity of the implemented policy value nor differentiation through the rounded terminal actor. With the same proposal and the same future policy, a different valid certificate changes information about cost, not the cost itself.

A cut obtained from the stored critic can be a useful proposal, but the tail defect in (33) must concern the original V_1. The same fixed feasible tail density supplies the all-action minorant; it is never differentiated as though it were an exact optimizer. The new service and the uniform-support conventional service both pay for tail proposals, original terminal evaluations, refused certificates, refinement, and fallback. Operation savings and actual returned-policy contrasts are reported separately.

7.3. *Paid precision and recursive use*

For any finite list of attempted resolutions, accepting at the first resolution whose outward intervals meet both original targets preserves qualification. Every unsuccessful earlier attempt remains part of the service cost. A geometric ladder alone gives no promise that a rejected action ever qualifies: its limiting supporting quadratic may remain too loose, or the action itself may be too costly. The unchanged fallback therefore remains part of the delivered policy.

For a declared fixed graph with exact-value quadrature allowance A/n^2, tail-localization allowance $B4^{-m}$, $A, B > 0$, positive costs $c_q n + c_b m$, and usable tolerance $\epsilon > 0$ after all other errors have been charged, an interior continuous optimum

satisfies

$$e_b = \frac{2c_b}{c_q\sqrt{A}\log 4}(\epsilon - e_b)^{3/2}, \qquad n = \sqrt{A/(\epsilon - e_b)}, \qquad m = \frac{\log(B/e_b)}{\log 4}. \tag{38}$$

The equation has one root in $(0,\epsilon)$. Domain boundaries such as $n \geq 1$ or $m \geq 0$ are checked separately; upward integer rounding preserves the allowance and adds less than $c_q + c_b$. This is an allocation for the stated work model, not an empirical compute price or a globally optimal discovery policy. Its proof is in the supplement.

At earlier dates the same witness construction can be used whenever the original continuation along the innovation has the required convexity and supplies global supports. Its local original-action gaps enter the retained dated policy account; constant certified continuation allowances propagate by discounting and cash invariance. Convexity is not asserted for an arbitrary learned network, an arbitrary nonlinear transition, or an unexecuted longer-horizon controller.

8. ACTION COMPARISONS THAT CAN CHANGE THE INVESTMENT DECISION

An accurate continuation approximation contributes economically when it helps choose a better feasible action at an acceptable complete cost. The retained R119 service experiments kept the implemented actions fixed. We now construct an action comparison whose acquisition error decreases with the distance between the actions, and use it to safeguard actual policy changes. The original nested AVaR criterion, information available to the economic agent, capacity constraint, and future controller remain those in Section 5.

At a fixed state write

$$f_a(z) = V_1(F(x,a,z)), \qquad Q_x(a) = S(x) + h(a) + \beta\rho_\alpha(f_a).$$

The actual complete-policy cost is $J_x(a)$, with the same implemented last-date policy for every current action. Its retained uniform guarantee is

$$J_x(a) = Q_x(a) + \eta_x(a), \qquad 0 \leq \eta_x(a) \leq \delta, \qquad \delta = \beta\varepsilon_1, \quad \varepsilon_1 = 2^{-20}. \tag{39}$$

The function η_x is not assumed convex or differentiable.

8.1. *A comparison modulus from the original technology*

The following constant concerns the original optimal continuation, not a fitted critic or an empirical gradient bound. This addition is restricted to the original two-dimensional application.

THEOREM 9—Original action comparison modulus: *For $d = 2$, $\beta = 15/16$, $R = 1/32$, and $\alpha \in \{1/2, 3/4\}$, put $t = 1 - \alpha$ and*

$$M_\alpha = \frac{1413}{128} + \beta \left\{ \frac{127}{16} + \frac{19625\alpha}{20480t} \right\}. \tag{40}$$

The original V_1 has an M_α-Lipschitz gradient on the state cube, with the continuous boundary extension. Consequently, for every original state and every two feasible actions,

$$\mathrm{Lip}_z(f_b - f_a) \le E_{ab} := \frac{4}{5} M_\alpha |b - a|. \tag{41}$$

Both conclusions include switches of the constrained terminal optimizer and of its active risk region.

The proof normalizes terminal investment as $A(y)v$, $v \in [0,1]$, and bounds the state semiconcavity of that fixed-domain objective. It therefore does not differentiate a rounded actor or ignore the capacity multiplier. The earlier expected-cost smoothness result in Section DB is retained with its original criterion; the extra term in (40) pays for the movement of the original AVaR tail.

Fix one feasible original-law tail density $q_u = t^{-1}\mathbf{1}_{[-R,u]\cup[u+2R\alpha,R]}$. Partition its nonempty tails into cells of lengths h_i, midpoints z_i, and weights $w_i = h_i/B$, where $B = 2Rt$. For each of the two actions, acquire valid point-value intervals at these midpoints and endpoints. Let L_a, L_b be their weighted midpoint lower sums and U_a, U_b their weighted endpoint-secant upper sums. Let $[M_-, M_+]$ enclose $\sum_i w_i\{f_b(z_i) - f_a(z_i)\}$ using the paired midpoint intervals. All of these acquisitions, including both action payloads, are paid.

THEOREM 10—A paid action-aligned original-policy certificate: *Define*

$$e_{ab} = \frac{E_{ab}}{4B} \sum_i h_i^2, \tag{42}$$

$$[d_-, d_+] = [L_b - U_a,\ U_b - L_a] \cap [M_- - e_{ab},\ M_+ + e_{ab}]. \tag{43}$$

This intersection contains $\mathbb{E}_P[q_u(f_b - f_a)]$. *If the two original tail deficits satisfy* $0 \le k_a(q_u) \le \kappa_a$ *and* $0 \le k_b(q_u) \le \kappa_b$, *the actual policy contrast satisfies*

$$J_x(b) - J_x(a) \in h(b) - h(a) + \beta[d_- - \kappa_a - \varepsilon_1, d_+ + \kappa_b + \varepsilon_1]. \tag{44}$$

Every numerical operation must enclose the displayed exact quantities outwardly. In particular, the separately acquired node errors and the tail defects are retained.

With exact point values, K-Lipschitz convex f_a, f_b, *an integer* $n \ge 1$, *and* $h_i \le 2Rt/n$, *the fixed-density interval has width at most*

$$\min\left\{\frac{2KRt}{n^2}, \frac{E_{ab}Rt}{n}\right\}. \tag{45}$$

For the original investment continuations one may take $K = 15$; *their additional smoothness sharpens the same computed interval to*

$$\min\left\{\frac{\min\{2KRt,\ 2M_\alpha R^2t^2\}}{n^2}, \frac{E_{ab}Rt}{n}\right\}. \tag{46}$$

The exactly identical realized action has contrast zero and requires no new density or point values.

The second interval in (43) uses a Lipschitz bound for a difference; it never treats that difference as convex. The common-density risk identity was frozen in R111 and proved in Theorem S.5. The present construction supplies its missing action-scaled integration bound and an executable refinement gate. Neither the identity nor generic safe policy iteration is claimed as a new neural property.

What the information bound does and does not establish. For a fixed known feasible tail density, let the information consist of exact point values of either function, K, an upper bound E on the Lipschitz constant of their difference, and the original uniform law. One scalar value of one function counts as one query; obtaining both values at a common point costs two. With this convention there is an algorithm using at most $4n + 8$ values and having the width in (45). Conversely, for every

deterministic adaptive algorithm using at most N such values, a known one-sided tail and a consistent pair force interval width at least

$$\min\left\{\frac{KRt}{48(N+1)^2}, \frac{ERt}{12(N+1)}\right\}. \tag{47}$$

Thus the worst-case order for this fixed-tail interface is $\min\{KRtN^{-2}, ERtN^{-1}\}$, for $N \geq 32$. Supplement Proposition S.8 gives the complete quantifiers and proof. Localization, original risk deficits, node precision, and descendant Bellman solves are additional information costs. The lower bound is not an oracle lower bound for an agent already given the algebraic formula or every additional smoothness restriction of the investment model.

8.2. *Safe refinement, stopping, and the future-policy floor*

Start from a completed original-policy incumbent. A proposed replacement is accepted only when the upper endpoint of (44) is strictly negative and its own policy interval passes the unchanged regret target $\tau = 1/256$ and width target $1/8192$. At the same state an original-optimum lower bound may be reused across actions; an action's policy interval may not. Refuse a candidate when its lower contrast endpoint is nonnegative, or retain the incumbent when the prespecified work limit is exhausted. Every refused trial remains in the complete cost.

PROPOSITION 11—Uniformly safe completed-policy replacement: *For every original state where the stated outward premises hold, any finite sequence of these replacements preserves feasibility, qualification, and weak improvement over its completed incumbent. If accepted contrast intervals are $[\underline{D}_j, \overline{D}_j]$, then the final original-policy gain satisfies*

$$0 \leq \sum_j (-\overline{D}_j) \leq J_x(a_{\text{initial}}) - J_x(a_{\text{final}}) \leq \tau. \tag{48}$$

A finite, measurable, statewise completed service satisfying these premises therefore defines a uniformly safe policy rule.

The proposed support directions and fractions $1, 1/2, 1/4$ are retained from R108–R110. The new comparison ladder changes which paid evidence can authorize their

realized action changes. A direct original-continuation optimizer receives the same comparison mechanism. The quadratic, polynomial, spline, grid, hinge, neural, and online representations receive the same original formulas and certificate access. The current experiment does not change the critic fitting objective or establish action-aware training convergence.

There is an explicit limit to refinement based only on (39). Even an exact $Q_x(b) - Q_x(a)$ leaves the sharp interface

$$J_x(b) - J_x(a) \in [Q_x(b) - Q_x(a) - \delta,\ Q_x(b) - Q_x(a) + \delta] \quad (a \neq b).$$

It does not uniformly authorize strict improvement when $Q_x(a) - Q_x(b) \leq \delta$. This is a restriction of the stated continuation-error information, not an assertion that the actual controller attains every endpoint or that more informative policy evaluation cannot help.

Action alignment also controls risk-witness mismatch. If $m_z = 181/64$, Theorem S.9 proves

$$k_a(q_u) \leq \left\{ \sqrt{\kappa_b} + E_{ab} \sqrt{\frac{\alpha}{2 m_z t}} \right\}^2. \tag{49}$$

At an exactly optimal candidate tail, the mismatch is quadratic in $|b - a|$. The implemented gate pays for and retains both direct deficits, and intersects the incumbent deficit with this transferred upper bound. This can tighten the lower contrast and refusal rule; the upper acceptance endpoint still uses the candidate deficit. No saved endpoint acquisition or upper-gate gain is attributed to that intersection. Supplement Proposition S.10 states the resulting finite stopping guarantee, including the tail, node, future, and qualification margins. A capped unresolved comparison is recorded as unresolved; it is not a certificate of optimality.

8.3. *From a safe decision rule to population economic gain*

A catalogue of successful gates is not by itself a population statement. Conditional on frozen actors and a fixed statewise service, let $X_1, \ldots, X_{64}$ be independent draws from the declared finite uniform initial-state law. Each complete service is applied separately to each state, so its action and gain do not depend on the other sampled

states. States may be shared across representations and risk levels. Let $G_j(x)$ be the gain over the corresponding completed incumbent, and let $[\ell_{ji}, u_{ji}]$ be saved, valid gain intervals contained in $[0, \tau]$.

PROPOSITION 12—Simultaneous transfer of certified economic gains: *Consider the fixed* 48 *gain groups and thresholds* $(\theta_1, \theta_2, \theta_3, \theta_4) = (2^{-20}, 2^{-18}, 2^{-16}, 2^{-14})$. *For each group and threshold count only* $\mathbf{1}_{\{\ell_{ji} \geq \theta_k\}}$. *Invert its binomial upper-tail probability at error* $1/7680$, *using an exact rational lower bracket endpoint, to obtain* p^-_{jk}. *Put* $r = 33\tau/128$ *and* $\theta_0 = 0$. *With conditional probability at least* 0.95, *simultaneously for all groups,*

$$\mathbb{E}G_j \geq \sum_{k=1}^{4} (\theta_k - \theta_{k-1}) p^-_{jk}, \tag{50}$$

$$\max\{0, \overline{\ell}_j - r\} \leq \mathbb{E}G_j \leq \min\{\tau, \overline{u}_j + r\}. \tag{51}$$

Neither independence across groups nor disjoint samples across risk levels are required.

The binomial inversion and bounded-mean concentration are classical (Clopper and Pearson, 1934, Hoeffding, 1963); the proof verifies their application to a uniformly safe original-policy gain, with all numerical interval uncertainty and the prespecified multiplicity account retained. The stored entropy bytes make the finite dyadic sample reproducible. Their independence and uniformity are the sampling assumption, not a mathematical conclusion established by a generator log. Positive gain alone does not establish adoption: acquisition, fitting, all attempted comparisons, inference, qualification, validation, and repeated decision service must be charged against that economic gain.

9. ORIGINAL-POLICY IMPROVEMENT AND COMPLETE ECONOMIC COST

9.1. *Frozen methods, original information, and independent states*

The economic primitives are exactly those in Section 5. The current experiment uses $d = 2$ and separately fixes $\alpha = 1/2$ and $3/4$. Both date-zero regret $1/256$ and policy interval width $1/8192$ must be certified. The terminal policy remains the original downward-40-bit controller, with its uniform error allowance $\varepsilon_1 = 2^{-20}$

TABLE I

FOUR PROSPECTIVELY FIXED COMPLETE DECISION SERVICES

Service	Original decision and paid additional work
Completed	The retained complete incumbent, including conventional and exact fallback when needed; no additional policy refinement.
Legacy safe	The unchanged retained three projected-support trials with fractions $1, 1/2, 1/4$ and separately qualified absolute policy intervals.
Paired	The same three support proposals, with new original paired contrast and transferred tail-defect intersection, using the fixed per-tail ladder $2, 4, 8, 16, 32, 64$.
Direct refine	A direct original-continuation proposal using 512 midpoint nodes and 24 derivative bisections, followed by the same new paid comparison and qualification ladder.

charged once in a complete-policy upper bound. Every risk integral concerns the original continuous uniform innovation distribution.

All learned critics use the same 256 original V_1 value intervals on a 16×16 midpoint state grid. Training targets are their midpoints. The representations are the online original solver, a six-feature quadratic critic, the retained ReLU critic, its exactly corresponding hinge representation, a coordinate spline, a full quadratic polynomial, a cubic polynomial, and bilinear conventional value-grid interpolation. The full learned architecture, feature construction, optimization trials and fitting checks are retained in the acquisition artifacts. The recursive proposal is computed from the represented continuation objective by the retained finite numerical search; original qualification is checked separately. No direct optimal-action labels or revised training target are introduced in this study. These last-date labels are acquired as $V_1 = T_1 g$ from the known terminal primitives, and their complete acquisition is paid. The numerical claims below concern this acquired Bellman step and the resulting date-zero policy corrections.

The last service is a strong traditional control; its numerical proposal is not asserted to be an exact optimizer. The online/direct-refine combination is also available without learned-label acquisition. Every service starts afresh and pays for

its own completed incumbent. Shared formulas and information access are identical across arms, while realized query counts and elapsed costs remain observable outcomes. The neural and equal-feature hinge pair provides a direct check on representation attribution.

The actual source and protocols are committed and pushed before acquisition. All acquired models and labels are then committed and pushed before drawing states. A single 640-byte OS-entropy acquisition encodes 128 independent 40-bit coordinate numerators. Its exact bytes and 64 states are separately committed before primary decisions. The declared population is the finite uniform law on $\{0, \ldots, 2^{40} - 1\}^2 / 2^{40}$, conditional on independent uniform entropy and its independence from the frozen actors. Common states across risks and methods improve comparability without creating additional independent observations. Each complete service is called on one initial state, so a batch fallback cannot change another sampled state's policy.

9.2. *Complete work and independently checked economic outcomes*

The primary census has $64 \times 2 \times 8 \times 4 = 4096$ decisions. Every tested candidate, refusal, exhausted ladder, density, queried terminal payload, final action and qualification interval is retained. Acquisition pays for labels, fits, candidate selection, fitting audits and serialization; the hinge arm pays for the ReLU feature acquisition it receives. Full cohort time adds cold loading, event-file setup, all singleton service calls and recording. The complete research ledger also reports work that cannot be assigned to a single deployment component. No favorable stage clock replaces a full-service clock. OpenBLAS, OMP and MKL use one thread; external load and processor frequency are not controlled.

Three fresh serial processes execute the fixed first 32 sampled states for all risks, arms and services. The service order follows the prospectively fixed rotation rule. All 6144 timing decisions and all clocks are kept. These repetitions describe timing dispersion and do not enter population gain inference as additional states. The primary service order, primary outcomes and all timing results remain available even if the order-specific medians are unfavorable.

Independent validation uses only rational economic primitives, with no import of the production compiler, interval arithmetic, fitting or risk-integration code. It covers every unique exact returned state/action/risk key with 64 whole-support cells, 129 distinct midpoint/endpoint nodes and 32 rational terminal bisections for an interior solution. Each terminal continuous-risk primal/dual identity is exact. Duplicated endpoints retain their correct multiplicities in the upper comparison distribution. The implemented future allowance is added once. Proposition S.12 adds an independently constructed original global optimum lower bound, allowing original regret and squared optimal-action distance to be bounded. All independently acquired work is charged separately; sharing a repeated validation key is not free input to any decision algorithm.

Population inference follows Proposition 12 with exactly 48 service-versus-own-incumbent gain groups and 192 fixed threshold events. The independently certified lower gains enter exact rational binomial inversion and the simultaneous staircase bound. The certified count is dominated by the latent count of true gains; this domination justifies conservative binomial inversion without assuming binomiality of the numerical certification indicator. Means use the fixed bounded-gain radius. All 64 observations remain in each group, including exact zero-action changes. An unavailable independent bound contributes the full safe interval rather than being omitted. Cross-representation economic contrasts are reported as independent numerical enclosures on this complete realized sample. They are not assigned an undeclared cross-representation population confidence level.

10. NEW ACTIONS UNDER THE ORIGINAL RECURSIVE OBJECTIVE

The prospectively frozen experiment changes root decisions while retaining the original continuous innovation law, risk operator, feasible actions and terminal controller. Each of 64 newly acquired two-coordinate states is evaluated at both risk levels by all eight representations and all four services. Training uses the same fixed value-label grid for every learned representation; the paired method changes the policy comparison and acquisition rule, not the critic's training target. The source, fitted actors and actual entropy/state artifacts were separately committed before

held-out decisions. Initial states are the independent units; shock nodes, risk levels, methods and timing repetitions add no independent observations.

We denote the completed incumbent by C, inherited compiled-safe improvement by L, the new paired rule by P, and the strong direct-refinement proposal with the same paired acceptance gate by D. Every returned root action is followed by the unchanged terminal controller. Independent rational bounds reconstruct the original policy objective J, with the single $\beta 2^{-20}$ implemented-future allowance; the original all-action lower yields regret and action-error upper bounds.

TABLE II: All-arm original-policy objectives and within-arm gains. C reports mean J; L/P/D report mean $J_C - J$ in 10^{-6} units. Bracket endpoints are displayed approximations of retained exact rational bounds.

α	Arm	C: mean J	L gain	P gain	D gain
1/2	Online	[0.642910, 0.642911]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]
1/2	Quadratic	[0.643090, 0.643092]	[178.836, 180.730]	[178.819, 180.709]	[179.318, 181.216]
1/2	ReLU	[0.643124, 0.643125]	[212.675, 213.885]	[212.675, 213.885]	[213.401, 214.591]
1/2	Hinge	[0.643124, 0.643125]	[212.675, 213.885]	[212.675, 213.885]	[213.401, 214.591]
1/2	Spline	[0.643251, 0.643253]	[340.325, 341.548]	[340.325, 341.548]	[340.621, 341.845]
1/2	Poly2	[0.643090, 0.643092]	[178.842, 180.736]	[178.824, 180.715]	[179.323, 181.221]
1/2	Poly3	[0.643145, 0.643147]	[233.603, 235.510]	[233.603, 235.510]	[234.410, 236.326]
1/2	Grid	[0.642928, 0.642930]	[17.819, 19.535]	[17.819, 19.597]	[17.996, 19.771]
3/4	Online	[0.654084, 0.654086]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]
3/4	Quadratic	[0.654278, 0.654280]	[192.865, 194.667]	[192.881, 194.728]	[193.363, 195.218]
3/4	ReLU	[0.654247, 0.654249]	[162.411, 163.501]	[162.411, 163.501]	[162.966, 164.055]
3/4	Hinge	[0.654247, 0.654249]	[162.411, 163.501]	[162.411, 163.501]	[162.966, 164.055]
3/4	Spline	[0.654477, 0.654479]	[391.623, 392.934]	[391.623, 392.934]	[392.135, 393.446]
3/4	Poly2	[0.654278, 0.654280]	[192.864, 194.666]	[192.881, 194.727]	[193.363, 195.217]
3/4	Poly3	[0.654314, 0.654316]	[228.740, 230.495]	[228.775, 230.529]	[229.556, 231.311]
3/4	Grid	[0.654127, 0.654129]	[42.198, 44.003]	[42.245, 44.050]	[42.486, 44.291]

Across the complete 4096-decision catalogue, 1170 returned decisions differ from their own completed incumbent and 2013 strict improvement steps are accepted. At the actual recorded resolutions, 2 accepted steps need the new mixed upper comparison: the separately valid convex-only scalar upper remains nonnegative. The transferred incumbent slack is needed for 0 recorded early non-improvement rejections. These counts describe which bound enables the recorded stopping de-

cision; they do not claim that a conventional method would fail at a later paid refinement or supply a measured counterfactual running time.

All 992 unordered same-risk method/service comparisons are retained: 244 strict negative, 202 strict positive, 31 exact zero and 515 intervals containing zero, in the declared A-minus-B order. Among the eight same-service quadratic-versus-ReLU comparisons, 1 favor quadratic, 1 favor ReLU and 6 are exact-zero or unresolved. These are comparisons of the recorded common-state sample. The ReLU and hinge actors use identical retained feature and coefficient arrays, with the full feature-acquisition charge paid by each; action identity is separately checked rather than inferred from that representation identity.

10.1. *Complete costs and adoption conditions*

TABLE III: Complete observed cost of serving all 64 original states, in seconds. Each entry includes its standalone acquisition charge and the actual full cohort clock, including rejected candidates and certificate work.

α	Arm	C	L	P	D
1/2	Online	1.887	7.341	11.674	6.838
1/2	Quadratic	2.144	6.319	7.220	4.441
1/2	ReLU	1.736	7.382	8.891	5.556
1/2	Hinge	1.736	5.737	9.357	5.591
1/2	Spline	1.734	5.758	10.004	5.514
1/2	Poly2	1.314	5.077	6.890	4.254
1/2	Poly3	1.301	5.644	7.182	4.665
1/2	Grid	1.439	5.799	9.004	4.539
3/4	Online	1.897	5.823	10.602	6.432
3/4	Quadratic	1.486	5.181	5.941	3.554
3/4	ReLU	1.827	5.815	7.701	5.426
3/4	Hinge	2.187	6.653	9.032	5.964
3/4	Spline	2.040	6.215	8.782	5.582
3/4	Poly2	1.384	5.273	6.645	3.739
3/4	Poly3	1.331	5.116	6.820	4.013
3/4	Grid	1.249	5.345	6.985	4.196

The 64 measured primary cohort clocks sum to 331.550 seconds. The separate independent rational validation process takes 1806.153 seconds; its full primitive trace and all per-policy costs are retained. This research validation cost is displayed

separately from a deployed service, which already pays its actual production certificate and every refused candidate. Acquisition charges include common labels, feature search and all fitting audits, serialization and conservatively assigned common overhead. They are allocations of measured components, not hypothetical independently timed fits. Three fresh serial timing processes serve the fixed first 32 states. All 192 cohort clocks and 6144 action comparisons are retained. Medians and ranges are descriptive; no timing confidence or throughput extrapolation is asserted.

The predeclared amortization scenarios use $B \in \{1, 64, 1024, 65536\}$ and $C_{m,r}(B) = (A_m + L_{m,r})/B + (T_{m,r} - L_{m,r})/32$. They reuse observed 32-state operating costs and are not measured scaling. For each original sample gain interval $[g_L, g_U]$ and extra cost ΔC, the reporter retains the exact nonnegative-price sets on which $g_L - \lambda \Delta C > 0$ and $g_U - \lambda \Delta C > 0$. Empty sets, zero or negative extra costs, and strict endpoints remain explicit. Comparison with the best conventional sample alternative intersects sufficient conditions against the full seven-arm, four-service menu, then across all three clocks. This is a conditional sufficient statement for the measured menu and cost scenario; it is not population adoption or a joint existence claim from overlapping upper bounds.

10.2. *The prespecified population gain family*

TABLE IV: Simultaneous population mean-gain bounds in 10^{-6} original-objective units for all 48 within-arm service comparisons. The fixed event and mean family has coverage at least 19/20 under the declared independent finite-uniform initial-state law, conditional on frozen actors and the ideal independent OS-entropy assumption.

α	Arm	L population gain	P population gain	D population gain
1/2	Online	[0.000, 1007.080]	[0.000, 1007.080]	[0.000, 1007.080]
1/2	Quadratic	[11.000, 1187.810]	[11.000, 1187.789]	[11.000, 1188.296]
1/2	ReLU	[6.603, 1220.965]	[6.603, 1220.965]	[6.603, 1221.671]
1/2	Hinge	[6.603, 1220.965]	[6.603, 1220.965]	[6.603, 1221.671]
1/2	Spline	[8.174, 1348.628]	[8.174, 1348.628]	[8.174, 1348.925]
1/2	Poly2	[11.000, 1187.816]	[11.000, 1187.795]	[11.000, 1188.301]
1/2	Poly3	[13.821, 1242.590]	[13.821, 1242.590]	[13.821, 1243.406]

Continued on next page

α	Arm	L population gain	P population gain	D population gain
1/2	Grid	[2.661, 1026.615]	[2.661, 1026.677]	[2.661, 1026.851]
3/4	Online	[0.000, 1007.080]	[0.000, 1007.080]	[0.000, 1007.080]
3/4	Quadratic	[11.314, 1201.747]	[11.314, 1201.808]	[11.314, 1202.298]
3/4	ReLU	[6.561, 1170.581]	[6.561, 1170.581]	[6.561, 1171.135]
3/4	Hinge	[6.561, 1170.581]	[6.561, 1170.581]	[6.561, 1171.135]
3/4	Spline	[8.378, 1400.014]	[8.378, 1400.014]	[8.378, 1400.526]
3/4	Poly2	[11.314, 1201.746]	[11.314, 1201.807]	[11.314, 1202.297]
3/4	Poly3	[13.472, 1237.575]	[13.472, 1237.609]	[14.192, 1238.391]
3/4	Grid	[6.436, 1051.083]	[6.436, 1051.130]	[6.436, 1051.371]

The exact binomial calculation uses all 64 states and the four fixed positive-gain thresholds. The bounded-mean calculation and event staircase use the same predeclared family allocation. No state is discarded for a small gain, broad interval, repeated draw or a conventional winner. Positive within-incumbent gain does not establish cross-representation superiority. The full development tables retain all original regret bounds, optimal-action-error upper bounds, precision failures, operation counters, timing repetitions, 992 sample contrasts and price domains. A reported execution pass asserts record consistency and finite compatibility; the associated theorems, rather than numerical pass counts, justify the certificate's mathematical scope.

10.3. *Economic gains and the strong conventional comparison*

The gains in Table II concern a concrete investment decision. Investment increases the two next-period stocks by $a/2$ and $a/4$, whereas the continuous innovation shifts them by $(z, -z)$. The innovation therefore changes their imbalance without changing their aggregate. The decision trades the convex current adjustment cost against stock tracking, the low-stock penalty, the capacity constraint and the original risk-sensitive continuation. At a common initial state, the current state cost cancels from the comparison. It is the action-dependent continuation difference, together with the adjustment-cost difference, that must justify an intervention. These are normalized original-objective units. The experiment supplies no monetary or consumption-equivalent welfare calibration.

Numerical displays in this discussion summarize retained exact rational intervals; the exact endpoints determine every sign and price condition. All seven learned representations improve on their own completed incumbents at both risk levels under each of L, P and D. The independent mean-gain lower bounds are strictly positive in all 42 such cells. For ReLU, P and L return the same policies: the mean original-policy gains are approximately $[212.675, 213.885] \times 10^{-6}$ at $\alpha = 1/2$ and $[162.411, 163.501] \times 10^{-6}$ at $\alpha = 3/4$. The corresponding independent mean-regret upper bounds fall from 215.906 to 2.626 and from 165.361 to 2.406, respectively, in the same 10^{-6} units. The original-action RMS-error upper bounds fall from 0.01470 to 0.001621 and from 0.01286 to 0.001551. These are bounds on error relative to an optimal original action; an exact optimal action was not supplied as an observed benchmark.

The magnitude of an improvement also depends on the initial actor's quality. The spline gains are larger than the ReLU gains because the completed spline incumbent has a larger original loss in this catalogue. They do not establish a superior final spline policy. More decisively, the online semi-analytic incumbent has a strictly lower sample mean original objective than every learned completed incumbent in all 14 comparisons. Every refinement leaves that online policy unchanged on the 64 states at both risks. Its independent mean-regret upper bounds are already 1.898×10^{-6} and 1.851×10^{-6}. Online also retains a strictly lower original objective than ReLU under L and P: the independently resolved online advantages are $[0.1506, 1.2864] \times 10^{-6}$ and $[0.06473, 1.0451] \times 10^{-6}$ at the two risks. ReLU with D reaches intervals overlapping the online policy, without resolving a strict mean advantage. All 512 ReLU–hinge policy pairs are exactly identical, and both representations pay the complete learned-feature acquisition charge. This control rules out a neural-specific policy advantage within the executed equal-feature comparison.

10.4. *What the new comparison changes*

The comparison with inherited improvement is more demanding than the comparison with an unrefined learned incumbent. P changes 391 of its 1024 returned actions and accepts 815 strict improvement steps. L changes 388 actions

and accepts 807 steps. Their final actions differ on only 12 of the 1024 common state/representation/risk cases. Nine of the 16 cell-level policy comparisons are exact identities; all seven remaining independent mean-value intervals contain zero. The retained intervals include both directions. Thus this experiment resolves gains over learned incumbents while leaving the incremental mean economic advantage of P over L unresolved.

The action-scaled bound does affect an actual stopping decision. At stored state index 29, $\alpha = 1/2$, the Poly3 candidate is accepted with $m = 4$: the new original-policy difference upper bound is approximately -1.05403×10^{-5}, whereas the separately valid convex-only scalar upper bound is 7.53783×10^{-6}. This is the same candidate in P and D, recorded twice; it is one economic example, not two independent observations. The mixed upper is active at 5961 of 9016 paid precision levels, but is necessary for acceptance at only those two recorded levels. The transferred incumbent tail-defect bound is inactive in all 2290 acquired density pairs. Its theorem remains valid; no observed reduction in incumbent tail work is attributed to it. These diagnostics establish use of the new comparison at a specified paid resolution. They do not establish failure of a traditional method after a further refinement or measure a counterfactual time saving.

The complete costs in Table III impose a further qualification. P costs more than L in every one of the 16 primary cells, with ratios from 1.143 to 1.821. It is also slower in all 48 matched fresh clocks; their complete-cost ratios range from 1.048 to 2.040. All three clocks, including their different magnitudes, remain reported. D is faster than P in every primary cell and all 48 matched clocks. Against L, D is faster in 41 clocks and slower in seven, with 14 favorable and two adverse cell medians. Its independent original-policy mean is strictly better than L and P in five cells—ReLU and hinge at both risks, and spline at $\alpha = 3/4$—with two exact online identities and nine unresolved comparisons. The conventional direct proposal is therefore an essential comparator to the new three-proposal rule.

Work counts and elapsed cost rank these procedures differently. Summed over the same 16 cells, L purchases 2,032,754 compiled terminal rows, P purchases 1,758,340, and D purchases 14,608,426. The corresponding sums of complete standalone 64-state costs are 94.478, 132.729 and 80.302 seconds. P retains 2257 rejected candidate

attempts, including 785 exhausted precision ladders; D retains 633 rejections, including 194 exhausted ladders. Identical-action shortcuts remain in these counts and are distinguished from purchased precision levels. The price of a point-information interface cannot be inferred from its point count alone. These measured results retain separate compiler, proposal, risk-query and certificate counters and the actual cohort clocks; they do not combine overlapping counters into a universal information unit or identify an unmeasured causal timing decomposition.

10.5. *Population scope and actual adoption conditions*

The frozen population analysis supplies a positive mean-gain lower bound for all 42 learned-incumbent comparisons, with the simultaneous event-and-mean family coverage stated in Table IV. The event-staircase lower bounds range from 2.661×10^{-6} to 14.192×10^{-6}. The ordinary bounded-mean lower bounds alone are zero; the predeclared positive-gain events supply the positive lower conclusions. The upper intervals remain broad. The six online comparisons have observed zero gains and lower bound zero; the finite sample does not prove that refinement is unnecessary everywhere. The inference concerns the declared finite-uniform initial-state distribution conditional on frozen actors and the ideal independent entropy assumption. It does not turn the 992 same-sample cross-representation contrasts into a new population comparison family.

A gain over a weak incumbent can pay for refinement without making its representation the preferred method. Using the actual 64-state workload and its own acquisition-plus-cohort cost, P improves the cost-augmented ReLU incumbent mean for computing prices in $[0, 0.001902379\ldots)$ at $\alpha = 1/2$ and $[0, 0.001769390\ldots)$ at $\alpha = 3/4$, in original-objective units per second. The exact rational open endpoints, not their printed decimal approximations, define these sets. Every one of the 32 ReLU comparisons against the full conventional menu, across the four predeclared amortization volumes and after intersection across all three clocks, has an empty sufficient-price set. The corresponding intersections of upper "not ruled out" conditions are empty as well. At the individual-clock level, 12 of 96 ReLU/menu upper-condition regions remain nonempty, all for direct refinement; the actual pri-

mary 64-state comparison has two such unresolved regions. These upper conditions establish no sufficient adoption region, but preserve the cases that a single timing scenario does not rule out. Across all representations, the 17 nonempty sufficient menu-price regions belong to completed conventional policies. These are conditions on the recorded sample and the declared cost scenario, not evidence of population adoption or measured throughput scaling. When a cheaper policy has a worse original objective, time savings can compensate only in the cost-augmented calculation; they do not improve the unpriced original Bellman objective.

The deployment comparison includes actual production certificates, refused proposals, acquisition, loading and cohort bookkeeping. The 64 primary cohort clocks sum to 331.550 seconds; the complete primary process, including three timing subprocesses and its other work, takes 841.861 seconds. The separate independent rational reconstruction takes 1806.153 seconds and is paid as research validation rather than silently assigned zero or amortized away in the deployment-price formula. Source/artifact checks, imports, process startup and final receipt work remain in the retained process and launcher ledger; they are not claimed as separately measured per-arm cohort components. Acquisition-plus-cohort sums charge standalone acquisition to each hypothetical deployment and therefore differ from the actual shared acquisition process. The mathematical progress is a valid action-dependent comparison and a certified rule that changes decisions under the original objective. The completed evidence identifies the remaining computational task precisely: preserve that validity while reducing paid decision work relative to strong conventional proposals and resolving advantages beyond ordinary equal-feature representations. Figure 1 displays every cell of the incremental P/L comparison, connecting the original-policy enclosures to all paid clocks.

11. CONCLUSION

Neural Bellman Operators connect learned continuation operators to feasible decisions and the economic value of their implemented future behavior. The construction requires a fixed original recursion and a complete accounting of the information used by its critic, actor and numerical certificate. The constrained

investment model makes these requirements explicit while retaining continuous innovations and nonlinear recursive risk.

The new action-comparison theory uses the geometry of the original Bellman problem. A constrained state-gradient modulus yields an action-scaled continuation-difference bound; convex integration and a constrained risk-defect transfer then provide computable comparison intervals. Qualification and a strict original-policy improvement bound jointly govern replacement of a completed incumbent. The proof also identifies what additional precision can resolve and what remains hidden behind a uniform future-policy allowance. This distinction gives a concrete stopping criterion and a specified information problem for subsequent NBO algorithms.

The independent experiment certifies gains over the completed learned incumbents under the original investment objective. All 42 learned-incumbent refinement groups have positive simultaneous population mean-gain lower bounds; the independent optimum supports also give much smaller upper bounds on original policy regret and root-mean-square optimal-action error. These gains coexist with a strong online incumbent that is already close to optimal. The new paired service changes twelve returned actions relative to inherited safe improvement, but no cell-level incremental mean advantage is resolved, and all sixteen primary costs and all forty-eight matched clocks are higher. The mixed comparison authorizes one shared candidate at a recorded resolution where the separate convex bound is inconclusive. The direct conventional proposal has a strictly lower independently enclosed sample mean cost than both L and P in five cells and is faster than P throughout the matched clocks. ReLU and hinge policies are identical. ReLU has empty sufficient full-menu adoption regions, and its upper-condition intersections across the three timing scenarios are empty; individual-clock upper conditions leave some cases unresolved. These findings preserve the independent neural contribution as an explicit scientific problem.

The remaining research priorities concern the joint design of continuation learning and paid economic decisions. They include acquisition losses that reduce action-comparison work, information about the actual future controller that resolves gains below the present allowance, and matched experiments with longer horizons, multiple controls and higher-dimensional continuous risk. The retained stochastic-

control, recursive-utility and stopped-boundary results provide separate rigorous foundations for this program. Progress on these problems must preserve the original objective, give conventional methods the same usable information, and demonstrate the economic value of the resulting policies together with their full computational cost.

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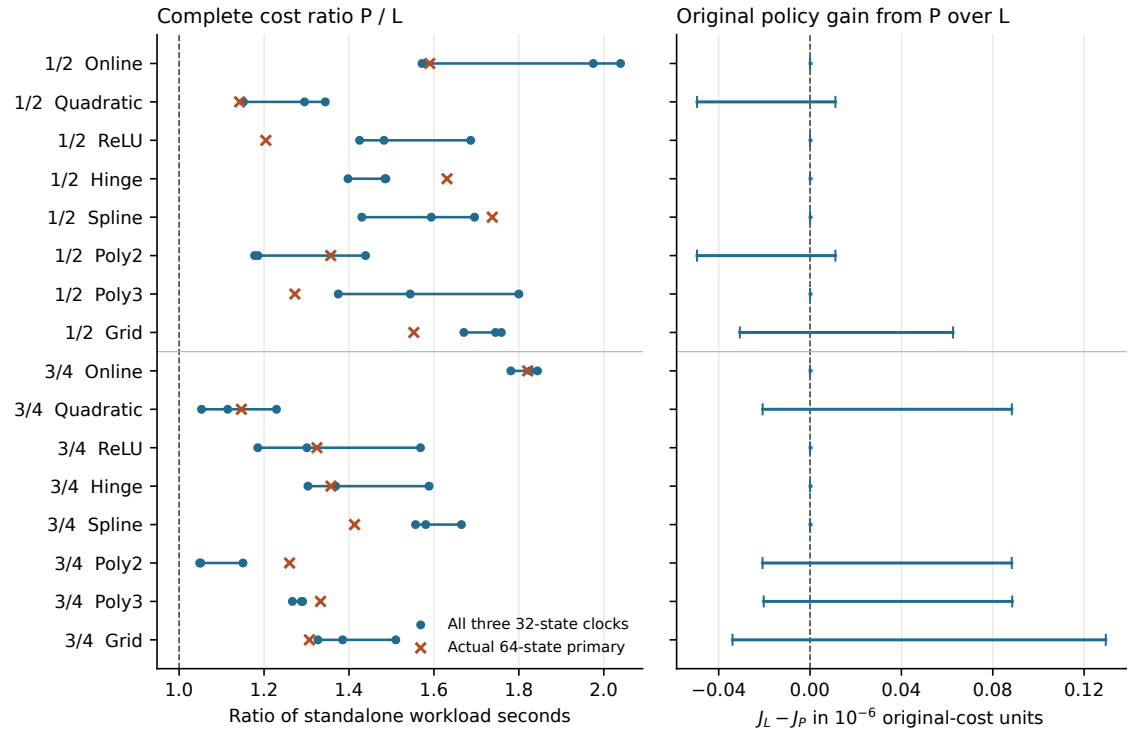


FIGURE 1.—Paired comparison versus inherited safe improvement in all sixteen representation–risk cells. The left panel displays the complete-cost ratio P/L from the primary 64-state cohort and each of the three matched 32-state process clocks. Its range connects observed clocks and is not a confidence interval. The right panel displays the independently reconstructed sample mean original-policy contrast $J_L - J_P$ in 10^{-6} original-cost units; positive values favor P. These are numerical enclosures on the common sample, not population confidence intervals. Every cell is included; nine contrasts are exact zero and the other seven contain zero. Exact rational values, all clocks, plotting code and figure hashes are retained with the source.